```
    this.root = new SsNode(false, children=[newChild1,
newChild2]    #4
```

#1 Method `insert` is defined on `SsTreee`. It takes a point and doesn't return anything.
#2 Call the insert function on the root, and store the result.
#3/#4 Only if the result of `insert` is not `null`, it needs to replace the old tree root with a newly created node, which will have the two nodes resulting from splitting the old root as its children.

10.3.3 Insertion: Variance, Means, and Projections

Now let's get into the details of the (many) helper methods we call in listing 10.6, starting with the heuristic method, described in listing 10.8, to find what's a node's closest child to a point `Z`: as mentioned, we will just cycle through a node's children, compute the distance between their centroids and `Z`, and choose the bounding envelope that minimizes it.

